DRUGS · ANTIBIOTICS: CEPHALOSPORINS AND CARBAPENEMS

Ceftazidime

Ceftazidime is the third-generation cephalosporin with reliable activity against Pseudomonas aeruginosa. In ENT it is used intravenously for necrotising otitis externa and skull base osteomyelitis, for Pseudomonas wound and cartilage infection that oral ciprofloxacin cannot treat, and in the immunocompromised patient. Its Gram-positive activity is weak.

1 CHAPTER ONE At a glance

The drug on one page.

1.1 Summary

CEFTAZIDIME

	SUMMARY
Class	Third-generation (anti-pseudomonal) cephalosporin
How it works	Cell wall synthesis inhibitor with an affinity for pseudomonal penicillin-binding protein 3
Spectrum	Pseudomonas aeruginosa, Enterobacterales, Haemophilus; poor against staphylococci and streptococci; not anaerobes, not MRSA, not enterococci
ENT uses	Necrotising otitis externa; Pseudomonas perichondritis with abscess or failure of oral treatment; Pseudomonas infection in cystic fibrosis sinus disease; febrile neutropenia
Adult dose	1 to 2 g every 8 hours intravenously (2 g every 8 hours in severe infection)
Child dose	25 to 50 mg/kg every 8 hours (maximum 6 g a day); up to 50 mg/kg every 8 hours in cystic fibrosis
Avoid	Beta-lactam anaphylaxis
Pregnancy	Use if needed

2 CHAPTER TWO How it works

Mechanism and handling.

2.1 Mechanism

Like other cephalosporins it blocks peptidoglycan cross-linking; its side chain gives it good penetration and affinity in Pseudomonas. Resistance arises from AmpC derepression, efflux and ESBL production; avibactam (ceftazidime-avibactam) restores activity against many resistant Gram-negatives and is reserved for specialist use.

2.2 Handling

Intravenous or intramuscular; renally excreted, so the dose falls with the eGFR.

3 CHAPTER THREE Using it in ENT

Indications and doses.

3.1 Necrotising otitis externa

Ceftazidime, or piperacillin-tazobactam, is the intravenous anti-pseudomonal choice, frequently with oral ciprofloxacin. After clinical response, falling CRP and ESR, and a resolving scan, therapy is continued orally with ciprofloxacin for a total of 6 weeks or more; recurrence is common in the first year. If cultures grow Aspergillus, add voriconazole.

3.2 Cartilage and wound infection

Pseudomonas perichondritis of the pinna with abscess, or not responding to oral ciprofloxacin, needs drainage and intravenous ceftazidime.

4 CHAPTER FOUR Safety

What goes wrong.

4.1 Adverse effects

- Diarrhoea and C. difficile; hypersensitivity.
- Neurotoxicity (encephalopathy, myoclonus, seizures) if the dose is not reduced in renal failure.
- Positive Coombs test, eosinophilia, raised liver enzymes.

4.2 Interactions

Aminoglycosides (additive nephrotoxicity; do not mix in one bag); chloramphenicol (antagonism).

5 CHAPTER FIVE For the examination

What the examiner wants to hear.

5.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- The intravenous and oral options for **Pseudomonas** in ENT.
- How you monitor treatment of necrotising otitis externa: pain, granulation, CRP and ESR, imaging (gallium or MRI).
- Why ceftazidime is not a good drug for streptococcal or staphylococcal infection.

5.2 Common pitfalls

COMMON PITFALLS

- Using ceftazidime as general broad cover in head and neck sepsis (it misses Gram-positives and anaerobes).
- Forgetting renal dose adjustment in elderly diabetics.

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